Lecture 2

Exercise 1

1 Preliminary

Note that $V_t = V(t)$ in Lecture 2.

Consider the following stochastic differential equation

$$\begin{cases} dV_t = rV_t dt + \sigma V_t dW_t, \\ V_0 = 1, \end{cases} \tag{1}$$

where r and σ are constant.

Let $f(t, V_t) = \ln V_t$. Then, we get

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial V_t} = \frac{1}{V_t}, \quad \frac{\partial^2 f}{\partial V_t^2} = -\frac{1}{V_t^2}.$$

Applying Itô Lemma yields

$$\begin{aligned} d\left(\ln V_t\right) &= d\left(f(t, V_t)\right) \\ &= \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial V_t} r V_t + \frac{1}{2}\frac{\partial^2 f}{\partial V_t^2}\sigma^2 V_t^2\right) dt + \frac{\partial f}{\partial V_t}\sigma V_t dW_t \\ &= \left\{0 + \frac{1}{V_t} r V_t + \frac{1}{2}\Big(-\frac{1}{V_t^2}\Big)\sigma^2 V_t^2\right\} dt + \frac{1}{V_t}\sigma V_t dW_t \\ &= \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t. \end{aligned} \tag{2}$$

Hence,

$$\ln V_T - \ln V_t = \int_t^T \left(r - \frac{1}{2}\sigma^2\right) ds + \int_t^T \sigma dW_s = \left(r - \frac{1}{2}\sigma^2\right)(T - t) + \sigma(W_T - W_t),$$

i.e.,

$$\ln\left(\frac{V_T}{V_t}\right) = \left(r - \frac{1}{2}\sigma^2\right)(T - t) + \sigma(W_T - W_t),$$

i.e.,

$$V_T = V_t \exp\left(\left(r - \frac{1}{2}\sigma^2\right)(T - t) + \sigma(W_T - W_t)\right). \tag{3}$$

Let $X_t = \ln V_t$. Then, it follows from (2) that we have

$$dX_t = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t. \tag{4}$$

Taking the expectation for (4) leads to

$$d(\mathbb{E}^*[X_t]) = \left(r - \frac{1}{2}\sigma^2\right) dt, \tag{5}$$

since $\mathbb{E}^*[dW_t] = 0$. Hence,

$$\mathbb{E}^*[X_T - X_t] = \mathbb{E}^*[X_T] - \mathbb{E}^*[X_t] = \int_t^T \left(r - \frac{1}{2}\sigma^2\right) ds = \left(r - \frac{1}{2}\sigma^2\right)(T-t). \tag{6}$$

Let $Y_t = X_t - \mathbb{E}[X_t]$. Then

$$\begin{aligned} dY_t &= dX_t - d(\mathbb{E}^*[X_t]) \\ &= \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t - \left(r - \frac{1}{2}\sigma^2\right) dt \\ &= \sigma dW_t. \end{aligned} \tag{7}$$

Hence,

$$\begin{aligned} \mathrm{Var}(X_T - X_t) &= \mathbb{E}^*[(X_T - X_t - \mathbb{E}^*[(X_T - X_t)])^2] \\ &= \mathbb{E}^*[(Y_T - Y_t)^2] \\ &= \mathbb{E}^*\left[\left(\int_t^T dY_s\right)^2\right] \\ &= \mathbb{E}^*\left[\left(\int_t^T \sigma dW_s\right)^2\right]. \end{aligned} \tag{8}$$

Applying Itô Isometry yields

$$\mathbb{E}^*\left[\left(\int_t^T \sigma dW_s\right)^2\right] = \mathbb{E}^*\left[\int_t^T \sigma^2 ds\right] = \sigma^2(T-t). \tag{9}$$

Hence,

$$\mathrm{Var}(X_T - X_t) = \sigma^2(T-t).$$

Therefore, we have

$$X_T - X_t \ \sim \ N\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t), \sigma^2(T-t)\right),$$

i.e.,

$$\ln V_T - \ln V_t \ \sim \ N\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t), \sigma^2(T-t)\right), \tag{10}$$

i.e.,

$$\ln\left(\frac{V_T}{V_t}\right) \ \sim \ N\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t), \sigma^2(T-t)\right),$$

i.e.,

$$\ln\left(\frac{V_T}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t) \;\sim\; N\left(0, \sigma^2(T-t)\right),$$

i.e.,

$$Z := \frac{\ln\left(\frac{V_T}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \;\sim\; N(0,1).$$

Hence, using the standard normal variable Z, we re-express V_T as

$$V_T = V_t \exp\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}Z\right). \tag{11}$$

2 Black-Scholes-Merton Geometric Brownian Models

Now let us consider

$$\begin{aligned}
P^*(V_T \le D) &= P^*\left(\frac{V_T}{V_t} \le \frac{D}{V_t}\right) \\
&= P^*\left\{\ln\left(\frac{V_T}{V_t}\right) \le \ln\left(\frac{D}{V_t}\right)\right\} \\
&= P^*\left\{\ln\left(\frac{V_T}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t) \le \ln\left(\frac{D}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)\right\} \\
&= P^*\left\{\frac{\ln\left(\frac{V_T}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \le \frac{\ln\left(\frac{D}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right\} \\
&= P(Z \le -d_2) \\
&= \Phi(-d_2),
\end{aligned} \tag{12}$$

where $d_2 = -\frac{\ln\left(\frac{D}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$.

At the beginning $t = 0$, we have

$$\begin{aligned}
P^*(V_T \le D) &= \Phi\left(\frac{\ln\left(\frac{D}{V_0}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-0)}{\sigma\sqrt{T-0}}\right) \\
&= \Phi\left(\frac{\ln D - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right).
\end{aligned} \tag{13}$$

Hence, the tax credit can be presented by

$$\begin{aligned} A(0) &= \mathbb{E}^*[e^{-rT}\kappa D \cdot I\{V(T) \geq D\}] \\ &= e^{-rT}\kappa D \cdot \mathbb{E}^*[I\{V(T) \geq D\}] \\ &= e^{-rT}\kappa D \cdot P^*(V(T) \geq D) \\ &= e^{-rT}\kappa D \cdot P^*(V_T \geq D) \\ &= e^{-rT}\kappa D \left(1 - P^*(V_T \leq D)\right) \\ &= e^{-rT}\kappa D \left(1 - \Phi\left(\frac{\ln D - \left(r - \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)\right). \end{aligned} \tag{14}$$

It follows from (10) that we have

$$\ln V_T \ \sim \ N\left(\left(r - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right), \tag{15}$$

for $t = 0$. Then, we have its density

$$p^*(V_T) = \frac{1}{\sqrt{2\pi T}\sigma V_T} \exp\left(-\frac{\left(\ln V_T - \left(r - \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right). \tag{16}$$

The bankruptcy cost is given by

$$\begin{aligned} C(0) &= \mathbb{E}^*[e^{-rT}(1-\alpha)V(T) \cdot I\{V(T) \leq D\}] \\ &= e^{-rT}(1-\alpha)\mathbb{E}^*[V(T) \cdot I\{V(T) \leq D\}] \\ &= e^{-rT}(1-\alpha)\mathbb{E}^*[V_T \cdot I\{V_T \leq D\}] \\ &= e^{-rT}(1-\alpha)\int_0^D V_T \cdot p^*(V_T) dV_T \\ &= e^{-rT}(1-\alpha)\int_0^D V_T \cdot \frac{1}{\sqrt{2\pi T}\sigma V_T} \exp\left(-\frac{\left(\ln V_T - \left(r - \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right) dV_T \\ &= e^{-rT}(1-\alpha)\int_0^D V_T \cdot \frac{1}{\sqrt{2\pi T}\sigma} \exp\left(-\frac{\left(\ln V_T - \left(r - \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right) d(\ln V_T) \\ &= e^{-rT}(1-\alpha)\int_0^D \exp(\ln V_T) \cdot \frac{1}{\sqrt{2\pi T}\sigma} \exp\left(-\frac{\left(\ln V_T - \left(r - \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right) d(\ln V_T) \\ &= e^{-rT}(1-\alpha)\int_0^D \frac{1}{\sqrt{2\pi T}\sigma} \exp\left(-\frac{\left(\ln V_T - \left(r - \frac{1}{2}\sigma^2\right)T\right)^2 - 2\sigma^2 T \ln V_T}{2\sigma^2 T}\right) d(\ln V_T) \\ &= e^{-rT}(1-\alpha)\int_0^D \frac{1}{\sqrt{2\pi T}\sigma} \exp\left(-\frac{\left(\ln V_T - \left(r + \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right) \exp(rT) d(\ln V_T) \\ &= (1-\alpha)\int_0^D \frac{1}{\sqrt{2\pi T}\sigma} \exp\left(-\frac{\left(\ln V_T - \left(r + \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right) d(\ln V_T) \\ &= (1-\alpha)\Phi\left(\frac{\ln D - \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right). \end{aligned} \tag{17}$$

A Itô Lemma

In its simplest form, Itô lemma states the following: for an Itô drift-diffusion process

$$dX_t = \mu_t dt + \sigma_t dW_t,$$

and any twice differentiable scalar function $f(t, x)$ of two real variables t and x, one has

$$df\left(t, X_t\right) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{\sigma_t^2}{2}\frac{\partial^2 f}{\partial x^2}\right) dt + \sigma_t \frac{\partial f}{\partial x} dW_t. \tag{18}$$

B Itô Isometry

The Itô isometry says that $X_t - X_s$ has variance

$$\mathbb{E}\left[\left(\int_s^t \xi_u dW_u\right)^2\right] = \mathbb{E}\left[\int_s^t \xi_u^2 du\right]. \tag{19}$$